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# *Bali-Phy* Tutorial

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## 1. Introduction

This tutorial contains step-by-step commands, assuming that you have already installed bali-phy. The manual [<http://www.bali-phy.org/README.html>] contains more detailed explanations for many things, so you might want to refer to it during the tutorial.

## 2. Setting up the ~/alignment\_files directory

Go to your home directory:

```
% cd ~
```

Make a directory called alignment\_files inside it:

```
% mkdir alignment_files
```

Go into the alignment\_files directory:

```
% cd alignment_files
```

Download the example alignment files:

```
% wget http://nucleus.biology.duke.edu/~bredelings/examples.tgz
```

Extract the compressed archive:

```
% tar -zxf examples.tgz
```

Take a look inside the examples directory:

```
% ls examples
```

Take a look at an input file:

```
% less examples/5S-rRNA/5d.fasta
```

Get some information about the alignment:

```
% alignment-info examples/5S-rRNA/5d.fasta
```

## 3. Command line options

What version of bali-phy are you running? When was it compiled? Which compiler? For what computer type?

```
% bali-phy -v
```

Look at the list of command line options:

```
% bali-phy --help
```

Look at them with the ability to scroll back:

```
% bali-phy --help | less
```

Some options have a short form which is a single letter:

```
% bali-phy -h | less
```

### 3.1. RNA

Analyze a data set, but don't begin MCMC. (This is useful to know if the analysis works, what model will be used, compute likelihoods, etc.)

```
% cd ~/alignment_files/examples
% bali-phy --show-only 5S-rRNA/5d.fasta
```

Finally, run an analysis! (This is just 10 iterations, so its not a real run.)

```
% bali-phy 5S-rRNA/5d.fasta --iterations=10
```

If you specify `--imodel=none`, then the alignment won't be estimated, and indels will be ignored (just like *MrBayes*).

```
% bali-phy 5S-rRNA/5d.fasta --iterations=10 --imodel=none
```

You can specify the alphabet, substitution model, insertion/deletion model, etc. Defaults are used if you don't specify.

```
% bali-phy 5S-rRNA/5d.fasta --iterations=10 --alphabet=DNA --smodel=TN --imodel=RS07
```

You can change this to the GTR, if you want:

```
% bali-phy 5S-rRNA/5d.fasta --iterations=10 --alphabet=DNA --smodel=GTR --imodel=RS07
```

You can add gamma[4]+INV rate heterogeneity:

```
% bali-phy 5S-rRNA/5d.fasta --iterations=10 --alphabet=DNA --smodel=GTR+gamma[4]+INV --imodel=RS07
```

### 3.2. Amino Acids

When the data set contains amino acids, the default is amino acids:

```
% bali-phy EF-Tu/12d.fasta --iterations=10
```

## 3.3. Codons

What alphabet is used here? What substitution model?

```
% bali-phy HIV/chain-2005/env-clustal-codons.fasta --show-only
```

What happens when trying to use the M0 model (e.g. positive selection)?

```
% bali-phy HIV/chain-2005/env-clustal-codons.fasta --show-only --smodel=M0
```

The M0 model requires a codon alphabet:

```
% bali-phy HIV/chain-2005/env-clustal-codons.fasta --show-only --smodel=M0 --alphabet=Codon
```

## 4. Output

See Section 6: Output [<http://www.bali-phy.org/README.html#output>] of the manual for more information about this section.

Try running an analysis with a few more iterations.

```
% bali-phy 5S-rRNA/25-muscle.fasta &
```

Run another copy of the same analysis:

```
% bali-phy 5S-rRNA/25-muscle.fasta &
```

You can take a look at your running jobs:

```
% jobs
```

### 4.1. Inspecting output files

Look at the directories that were created to store the output files:

```
% ls
% ls 25-muscle-1/
% ls 25-muscle-2/
```

See how many iterations have been completed so far:

```
% wc -l 25-muscle-1/C1.p 25-muscle-2/C1.p
```

Wait a second, and repeat the command.

```
% wc -l 25-muscle-1/C1.p 25-muscle-2/C1.p
```

See if you can determine the following information from the beginning of the C1.out file: What command was run? When was it run? Which computer was it run on? Which directory was it run in? Which directory contains the output files? What was the process id (PID) of the running bali-phy program? What random seed was used? What was the input file? What alphabet was used to read in the file? What substitution model was used to analyze the file? What insertion/deletion model was used to analyze the file?

```
% less 25-muscle-1/C1.out
```

Examine the file containing the sampled trees:

```
% less 25-muscle-1/C1.trees
```

Examine the file containing the sampled alignments:

```
% less 25-muscle-1/C1.P1.fastas
```

Examine the file containing the successive best alignment/tree pairs visited:

```
% less 25-muscle-1/C1.MAP
```

## 4.2. Summarizing the output

Try summarizing the sampled numerical parameters (e.g. not trees and alignments):

```
% statreport --help
% statreport 25-muscle-1/C1.p 25-muscle-2/C1.p > Report
% statreport 25-muscle-1/C1.p 25-muscle-2/C1.p --mean > Report
% statreport 25-muscle-1/C1.p 25-muscle-2/C1.p --mean --median > Report
% less Report
```

Now let's examine the summaries using a graphical program. If you are using Windows or Mac, run Tracer, and press the + button to add these files. What kind of ESS do you get? If you are using Linux, do

```
% tracer 25-muscle-1/C1.p 25-muscle-2/C1.p &
```

Now let's compute the consensus tree for these two runs:

```
% trees-consensus --help
% trees-consensus 25-muscle-1/C1.trees 25-muscle-2/C1.trees > c50.PP.tree
% trees-consensus 25-muscle-1/C1.trees 25-muscle-2/C1.trees --report=consensus > c50.PP.tree
% less consensus
% figtree c50.PP.tree &
```

Now let's see if there is evidence that the two runs have not converged yet:

```
% trees-bootstrap --help
% trees-bootstrap 25-muscle-1/C1.trees 25-muscle-2/C1.trees > partitions.bs
% less partitions.bs
```

Now let's use the analysis script to run all the summaries and make a report:

```
% bp-analyze.pl 25-muscle-1/ 25-muscle-2/
% firefox Results/index.html &
```

## 5. Starting and stopping the program

We didn't specify the number of iterations to run in the section above, so the two analyses will run for 100,000 iterations, or until you stop them yourself. See Section 8: Convergence and Mixing: Is it done yet? [[http://www.bali-phy.org/README.html#mixing\\_and\\_convergence](http://www.bali-phy.org/README.html#mixing_and_convergence)] of the manual for more information about when to stop an analysis.

In order to stop a running job, you need to kill it. One way of stopping bali-phy analyses is this:

```
% killall bali-phy
```

However, beware: if you are running multiple analyses, this will terminate all of them.

## 6. Multi-gene analyses

In this section we'll practice running analysis with multiple partitions. Dividing the data into multiple partitions is useful because different partitions can have different models, or can have different parameters for the same model. This is described in more detail in section 4.3 of the manual [<http://www.bali-phy.org/README.html>].

## 6.1. A simple multi-gene analysis

Let's look at a data set that is divided into three partitions:

```
% alignment-info ITS/ITS1-trimmed.fasta
% alignment-info ITS/5.8S.fasta
% alignment-info ITS/ITS2-trimmed.fasta
```

### 6.1.1. Running the analysis

We can run an analysis of this partitioned data simply by supplying a number of different alignment files as input to bali-phy. Let's run an analysis of these three alignment files:

```
% bali-phy ITS/ITS1-trimmed.fasta ITS/5.8S.fasta ITS/ITS2-trimmed.fasta --smodel=TN --imodel=RS07
% bali-phy ITS/ITS1-trimmed.fasta ITS/5.8S.fasta ITS/ITS2-trimmed.fasta --smodel=TN --imodel=RS07
```

You could leave off the `--smodel=TN --imodel=RS07` part of the command line:

```
% bali-phy ITS/ITS1-trimmed.fasta ITS/5.8S.fasta ITS/ITS2-trimmed.fasta &
```

This would give the same output, since TN and RS07 are the defaults.

### 6.1.2. What did the analysis do?

Now, let's look at sampled continuous parameters:

```
% statreport ITS1-trimmed-5.8S-ITS2-trimmed-1/C1.p | less
% tracer ITS1-trimmed-5.8S-ITS2-trimmed-1/C1.p &
```

You'll see that each partition has a TN (Tamura-Nei) substitution model, as well as an RS07 indel model. Each partition has its own copy of the TN parameters and the RS07 parameters.

### 6.1.3. Question

The partitions share a common tree shape, including the same relative branch lengths. However, the size of the tree for each partition is different. We scale the whole shared tree by  $\mu_1$  in partition 1,  $\mu_2$  in partition 2, etc. The  $\mu$  parameters give the average branch length in that partition. Thus, partitions with a smaller  $\mu$  value have slower evolution.

Do the different partitions of this data set have the same evolutionary rates? Do the different partitions of this data set have the same base frequencies?

## 6.2. Using different models in different partitions

### 6.2.1. Using different substitution models

Now let's try to specify different models for different partitions. Here we've used a command-line trick with curly braces `{}` to avoid typing some things multiple times.

```
% bali-phy ITS/{ITS1-trimmed,5.8S,ITS2-trimmed}.fasta --smodel=1:GTR --smodel=2:HKY --smodel=3:GTR
```

We've also specified different substitution models for each partition. Take a look at the C1.p file for this analysis to see what parameters appear.

## 6.3. Using different indel models

We can also specify different indel models for each partition:

```
% bali-phy ITS/{ITS1-trimmed,5.8S,ITS2-trimmed}.fasta --imodel=1:RS07 --imodel=2:none --imodel=3:RS07
```

There are only two indel models: RS07, and none. Specifying `--imodel=none` removes the insertion-deletion model and parameters for a partition. It also disables alignment estimation for that partition.

## 6.4. Sharing model parameter between partitions

We can also specify that some partitions with the same model also share the same parameters for the model:

```
% bali-phy ITS/{ITS1-trimmed,5.8S,ITS2-trimmed}.fasta --smodel=1,3:GTR --imodel=1,3:RS07 --sm
```

This means that the information is pooled between the partitions to infer the shared parameters.

## 6.5. Sharing substitution rates between partitions

We can also specify that some partitions with the same model also share the same parameters for the model:

```
% bali-phy ITS/{ITS1-trimmed,5.8S,ITS2-trimmed}.fasta --smodel=1,3:GTR --imodel=1,3:RS07 --sm
```

This means that the branch lengths for partitions 1 and 3 are the same, instead of being independently estimated.

## 7. Option files

You can collect command line options into a file for later use. Make a text file called `analysis1.script`:

```
align = ITS/ITS1-trimmed.fasta
align = ITS/5.8S.fasta
align = ITS/ITS2-trimmed.fasta
smodel = 1,3:TN+DP[3]
smodel = 2:TN
imodel = none
same-scale = 1,3:mu1
```

You can run the analysis like this:

```
% bali-phy -c analysis1.script --show-only
```